Pearls AQI Predictor

End-to-End Air Quality Forecasting System using Hopsworks, Machine Learning, and Streamlit

Submitted By

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Internship Project

10Pearls Shine Internship Program

(Data Sciences)

Abstract

Air pollution has become a major environmental and public health concern worldwide. The objective of this project is to design and implement an end-to-end Air Quality Index (AQI) forecasting system capable of predicting future air quality conditions using machine learning techniques and a modern MLOps architecture.

The developed solution automatically collects environmental and pollutant data, performs feature engineering, stores data in a Feature Store, trains multiple machine learning models, registers the best-performing model in a Model Registry, and serves predictions through an interactive Streamlit dashboard. The system includes explainability using SHAP, automated CI/CD pipelines using GitHub Actions, AQI alert generation, and 72-hour forecasting capabilities.

The final system achieved an R^2 score of 0.9389 using a Gradient Boosting Regressor model and was successfully deployed as a cloud-hosted Streamlit application.

1. Introduction

Air Quality Index (AQI) is a numerical representation of air pollution levels and their potential impact on human health. Accurate AQI forecasting can help governments, organizations, and citizens make informed decisions regarding outdoor activities and environmental planning.

The goal of this project was to build a complete machine learning pipeline capable of:

- Collecting AQI and weather data
- Engineering meaningful forecasting features
- Training and evaluating machine learning models
- Automating feature and training pipelines
- Providing explainable predictions
- Visualizing results through a user-friendly dashboard

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2. Project Objectives

The primary objectives of the project were:

- Build an automated AQI forecasting pipeline
- Store engineered features in Hopsworks Feature Store
- Train multiple machine learning models
- Evaluate models using MAE, RMSE, and R^2
- Register trained models in Hopsworks Model Registry
- Automate pipelines through GitHub Actions
- Develop an interactive Streamlit dashboard
- Implement SHAP explainability
- Generate AQI health alerts
- Deploy the application to the cloud

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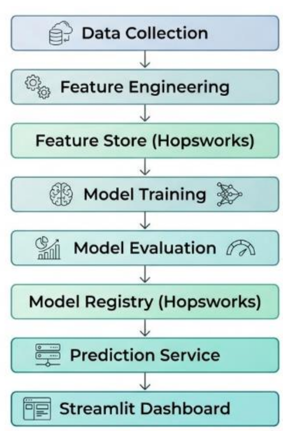
3. System Architecture

The overall architecture consists of the following stages:



The architecture follows modern MLOps principles by separating feature management, model management, automation, and deployment.

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4. Data Collection

The system retrieves historical and live environmental information from Open-Meteo APIs.

Collected variables include:

- AQI
- PM2.5
- PM10
- CO
- NO_2
- O_3
- Temperature
- Humidity
- Pressure
- Wind Speed

Historical backfilling was performed to generate sufficient training data for forecasting.

```
timestamp,temperature,humidity,wind_speed,pressure,pm25,pm10,co,no2,o3,aqi
2026-02-08 19:00:00,20.35,89.69294,9.761578,1013.4741,33.5,46.9,526.0,33.6,45.0,37.92
2026-02-08 20:00:00,20.15,88.838524,8.951267,1013.2736,32.4,46.8,419.0,27.4,49.0,37.02
2026-02-08 21:00:00,20.0,88.54785,7.989242,1012.97327,32.1,48.5,336.0,22.2,51.0,36.73
2026-02-08 22:00:00,19.55,90.1991,8.43699,1012.5723,32.8,53.6,260.0,17.4,53.0,37.68
2026-02-08 23:00:00,19.4,90.47284,7.289445,1011.9724,32.5,55.3,209.0,13.6,54.0,37.45
2026-02-09 00:00:00,18.65,93.61844,4.3573384,1011.87036,34.6,61.2,257.0,14.0,50.0,39.14
2026-02-09 01:00:00,18.75,92.74196,4.8965297,1011.8707,34.8,60.5,522.0,21.5,40.0,38.72
2026-02-09 02:00:00,18.8,91.58149,5.414979,1012.5704,38.8,65.7,887.0,31.4,27.0,41.3
2026-02-09 03:00:00,19.5,87.39624,7.596341,1013.37164,47.6,75.0,1071.0,35.9,24.0,47.78
2026-02-09 04:00:00,21.55,78.24292,7.2044983,1014.0768,46.6,77.1,891.0,30.0,41.0,49.37
2026-02-09 05:00:00,23.55,61.41637,12.108261,1014.782,37.4,79.2,531.0,18.7,67.0,47.4
2026-02-09 06:00:00,24.95,55.042286,14.404499,1014.6858,34.1,74.7,260.0,9.7,89.0,46.8
2026-02-09 07:00:00,25.4,53.759243,15.661035,1014.2871,31.2,67.5,196.0,6.0,100.0,45.0
2026-02-09 08:00:00,25.55,53.282288,16.036171,1013.48846,28.4,64.3,221.0,4.5,106.0,43.64
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5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset.

Key analyses included:

- AQI distribution
- Hourly AQI trends

- Monthly AQI patterns
- Weekday vs Weekend comparison
- Pollutant correlation analysis
- Weather impact analysis

The EDA revealed strong relationships between AQI and pollutant concentrations, particularly PM2.5 and PM10.

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6. Feature Engineering

Several engineered features were created to improve forecasting performance.

Time Features

- Hour
- Day
- Month
- Day of Week
- Weekend Indicator
- Rush Hour Indicator

Lag Features

- previous_aqi
- aqi_lag_3
- aqi_lag_6
- aqi_lag_12

Rolling Statistics

- rolling_avg_3
- rolling_avg_6
- rolling_avg_24

Trend Features

- aqi_change
- aqi_trend

Pollution Features

- pollution_index

Target Variable

The forecasting target was defined as:

future_aqi = aqi.shift(-72)

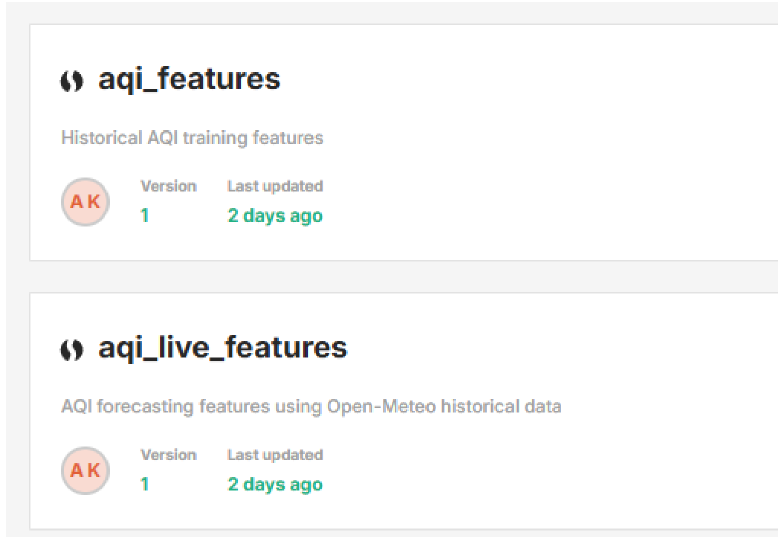
This allows prediction of AQI levels 72 hours into the future.

7. Feature Store Implementation

Hopsworks Feature Store was used to store and manage engineered features.

Feature Groups:

- aqi_training_features
- aqi_live_features



Benefits:

- Centralized feature management
- Reusability across pipelines
- Consistent training and inference data
- Scalable architecture

8. Machine Learning Models

Three machine learning models were trained and evaluated.

Ridge Regression

A linear regression model with regularization.

Random Forest Regressor

An ensemble learning model based on multiple decision trees.

Gradient Boosting Regressor

An advanced boosting model that sequentially improves prediction performance.

9. Model Evaluation

The models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

Results

Model	MAE	RMSE	R^2
Ridge	1.7785	5.0609	0.9179
Random Forest	1.9487	4.7165	0.9287
Gradient Boosting	1.8153	4.3660	0.9389

Best Performing Model

Gradient Boosting achieved the highest R^2 score:

$R^2 = 0.9389$

and was selected as the production model.

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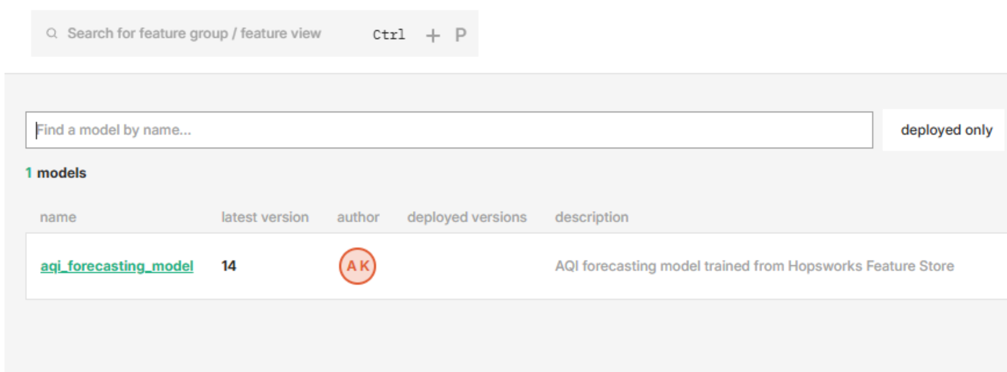
10. Model Registry

The trained model was stored in the Hopsworks Model Registry.

Benefits include:

- Model versioning
- Reproducibility
- Deployment readiness
- Centralized model management



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11. Forecasting System

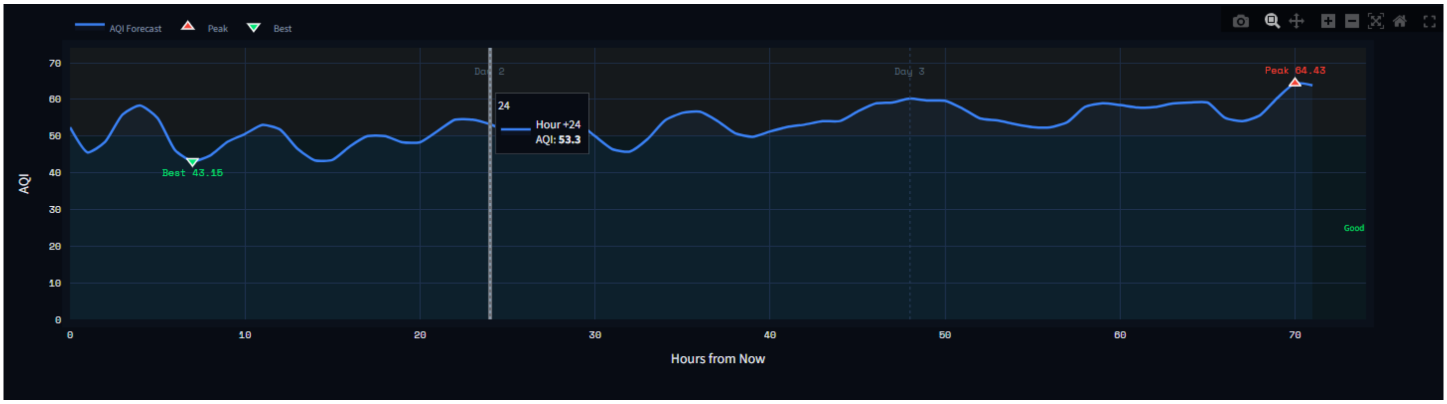
The system generates AQI forecasts for the next 72 hours.

Features:

- Current AQI display
- Peak AQI detection
- Best AQI period identification

- Hourly forecast visualization

The forecasting module uses autoregressive prediction techniques based on the latest available features.



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12. Explainability Using SHAP

To increase transparency and trust in model predictions, SHAP (SHapley Additive exPlanations) was integrated.

Implemented visualizations:

- Feature Importance Plot
- SHAP Waterfall Chart
- Top Driver Identification
- Feature Impact Breakdown

These explanations help users understand why the model predicts a specific AQI value.

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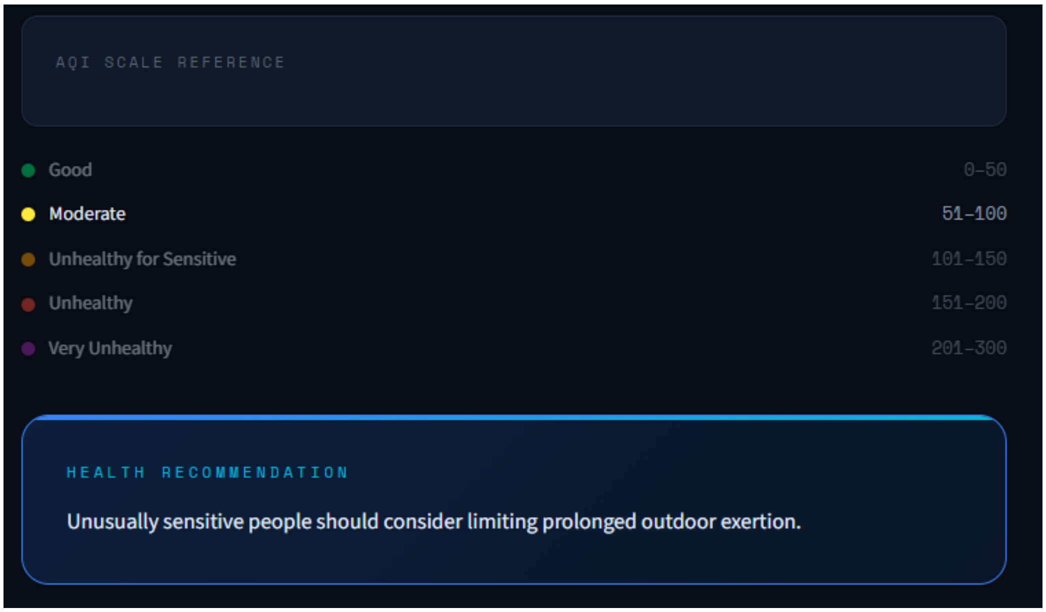
13. AQI Alert System

A rule-based alert system was implemented.

Alert Levels:

- Good
- Moderate
- Unhealthy for Sensitive Groups
- Unhealthy
- Very Unhealthy
- Hazardous

The dashboard automatically displays health recommendations according to the predicted AQI level.



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14. CI/CD Automation

GitHub Actions was used to automate the pipelines.

Feature Pipeline

Runs automatically every hour.

Responsibilities:

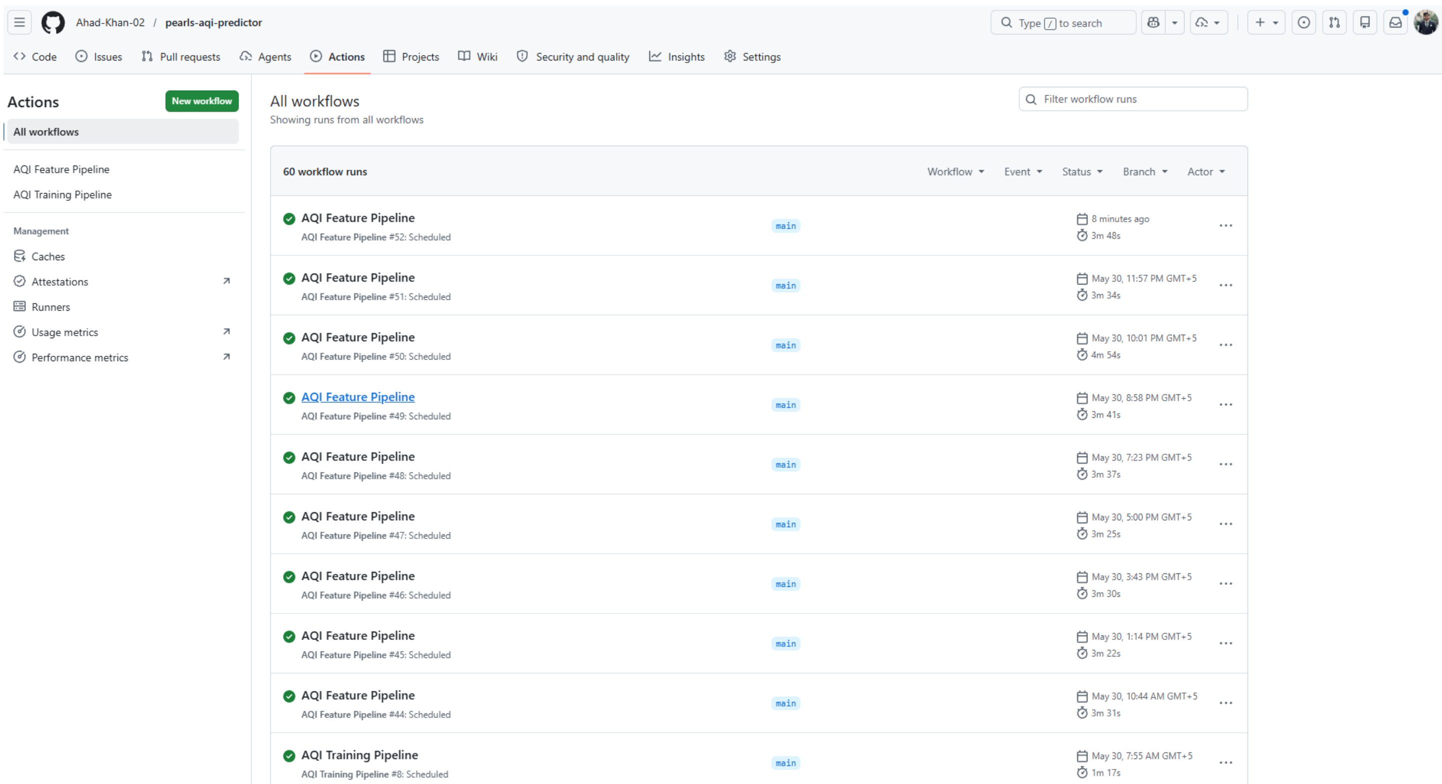
- Fetch latest environmental data
- Generate features
- Update Feature Store

Training Pipeline

Runs automatically every day.

Responsibilities:

- Load historical features
- Train models
- Evaluate performance
- Update Model Registry



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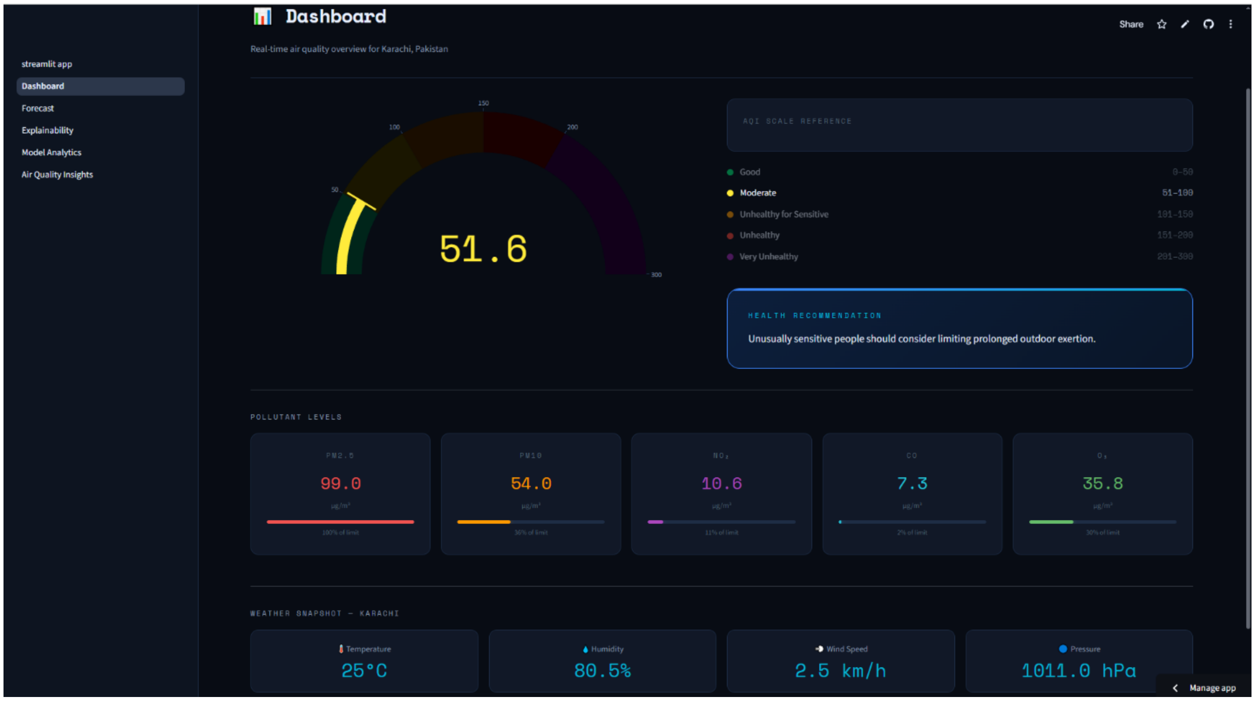
15. Dashboard Implementation

The dashboard was developed using Streamlit.

Modules include:

Dashboard

- Live AQI
- AQI Gauge
- Pollutant Monitoring
- Weather Snapshot



Forecast

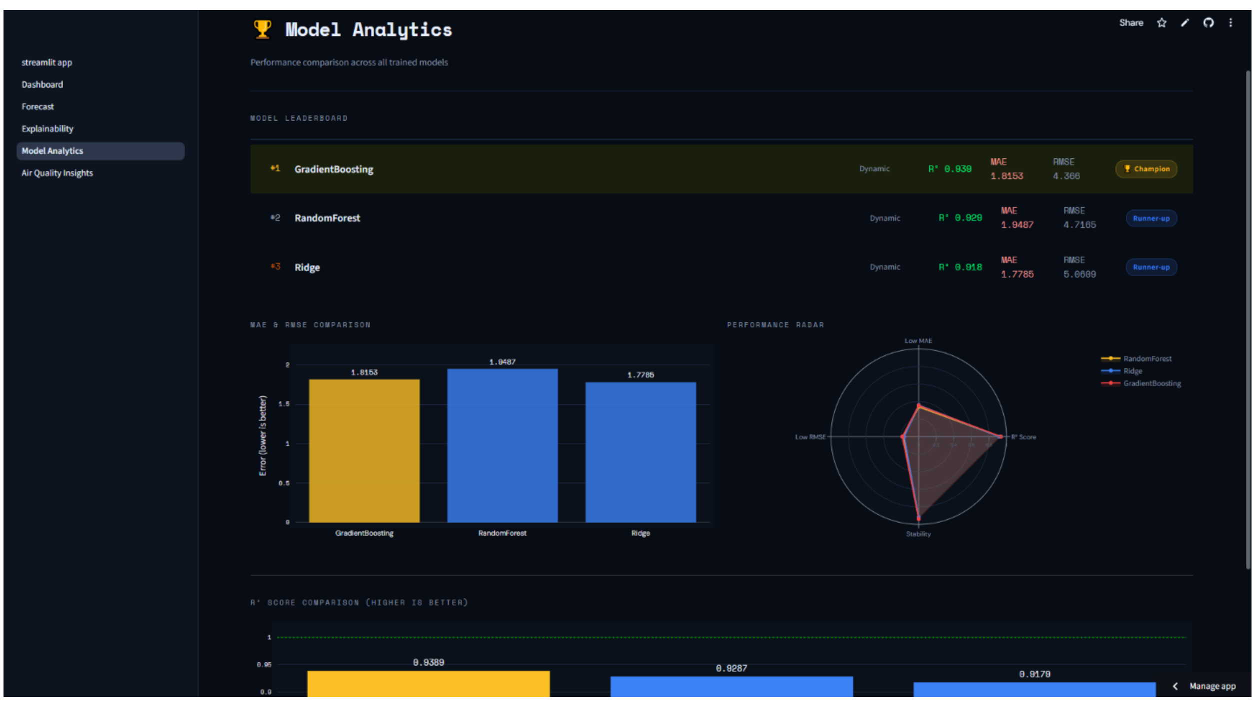
- 72-Hour AQI Prediction
- Peak and Best AQI Indicators

Explainability

- SHAP Feature Analysis
- Prediction Drivers

Model Analytics

- Model Leaderboard
- MAE Comparison
- RMSE Comparison
- R² Comparison



Air Quality Insights

- AQI Trends
- Correlation Analysis
- Statistical Insights

16. Deployment

The application was deployed using Streamlit Cloud.

Deployment URL:

https://pearls-aqi-predictor-1.streamlit.app/

Source Code Repository:

https://github.com/Ahad-Khan-02/pearls-aqi-predictor

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17. Technologies Used

Component	Technology
Programming Language	Python
Data Source	Open-Meteo API
Data Processing	Pandas, NumPy
Visualization	Plotly, Matplotlib
Feature Store	Hopsworks
Model Registry	Hopsworks
Machine Learning Models	Ridge Regression, Random Forest, Gradient Boosting
Explainability	SHAP
Dashboard	Streamlit
CI/CD Automation	GitHub Actions
Deployment	Streamlit Cloud
Version Control	Git & GitHub

18. Challenges Faced

Several challenges were encountered during development:

- Time-series data leakage during training
- Hopsworks schema compatibility issues
- Feature Store synchronization problems
- TensorFlow compatibility challenges
- Forecast realism and stability
- Model registry integration

These issues were systematically resolved through iterative debugging and redesign.

19. Future Improvements

Potential future enhancements include:

- Integration of deep learning models (PyTorch/LSTM)
- Multi-city forecasting support
- Real-time notification system
- Mobile application integration
- Advanced ensemble forecasting
- Weather-aware forecasting models

20. Conclusion

This project successfully developed a complete end-to-end AQI forecasting platform using modern machine learning and MLOps practices.

The system automates data collection, feature engineering, model training, model management, forecasting, explainability, and deployment within a scalable architecture.

The final Gradient Boosting model achieved an R^2 score of 0.9389, demonstrating strong forecasting performance. The deployed Streamlit application provides users with real-time AQI monitoring, 72-hour forecasts, explainable predictions, and health alerts.

The project satisfies all major requirements of the internship challenge while providing a practical and extensible solution for air quality forecasting.